



MDEP-CE Cohort 7

Hackathon 2026

SAP Institute for Product and Engineering
August 2026

Agenda

- 01 Welcome & Hackathon Overview**
Introduction to the hackathon goals, objectives, and expectations

- 02 Use Case Introduction**
Overview of the maintenance scenario and business context

- 03 Business Challenge**
Understanding the need for transforming maintenance planning for greater efficiency.

- 04 Time & Deliverables**
Review project milestones, key activities, and expected outcomes.

- 05 Solution Approach**
Explore how SAP AI technologies can enable an intelligent solution.

- 06 Q&A**
Address questions and discuss next steps.

Hackathon Challenge

In manufacturing industries, timely maintenance is critical to keeping production running smoothly and minimizing costly downtime.

Maintenance supervisor must verify that all required spare parts and materials are available to complete the maintenance order. Today, this involves switching between inventory systems and spreadsheets, manually validating each item.

Hackathon Objective

Develop an AI-powered solution that transforms the maintenance planning process. By analyzing a maintenance order, the solution should validate material availability, identify shortages, and recommend next steps through a single intelligent interface enabling faster and smarter maintenance decisions.



Maintenance Supervisor Challenges

TO

AI Assistant Agent Benefits



- ◆ Multiple Systems >
- ◆ Manual spare parts check >
- ◆ Time-consuming validation >
- ◆ Missing parts >
- ◆ Poor inventory visibility >



- ◆ One unified view >
- ◆ Instant availability check >
- ◆ Automated validation >
- ◆ Early shortage alerts >
- ◆ Real-time inventory insights >

Three-Week Project Plan

Week	Focus Area	Key Activities
Week 1	Discovery & Planning	Understand the use case and business requirements, design the application prototype. Finalize and share solution approach with your coach.
Week 2	Solution Design & Technical Preparation	Define your solution architecture and identify the services and agentic skills needed.
Week 3	Development & Demo Preparation	Develop the agentic AI solution, build an overview dashboard for your use case, and prepare the live end-to-end demo and presentation.

Teams and Coaches

Team	Team Name	Room	Coach
Team 1	One Vision	Jackson A	Taha
Team 2	BRICKS	Jackson B	Prashant
Team 3	JASMN Fusion	Jackson D	Kanthi
Team 4	Hexcore	Lovelace A	Taha
Team 5	Cactus	Lovelace D	Poorna



AI MAINTENANCE PLANNER

Driving Performance. Powering Progress



AI Maintenance Planner Solution

An AI-powered solution that ensures maintenance orders are execution ready by validating material availability and providing actionable recommendations.

Solution Approach:

- Develop a CAP-based backend service running on **SAP BTP** to expose maintenance and inventory data.
- Build agentic AI solution using **SAP Joule Studio** that assists maintenance supervisors.
- Leverage **SAP BDC** to model data and build an overview dashboard that provides maintenance planners with actionable insights.

Scenario 1: Agentic AI for Maintenance Order Insights

Objective

Build an AI-driven Maintenance Order Readiness and Execution Agent for an automotive plant that minimizes unplanned downtime and total maintenance cost by making faster, better decisions across the entire maintenance order lifecycle, from prioritization and material readiness to execution progress and completion ETA.

Problem (Supervisor's voice)

"I'm flooded with maintenance orders coming from different sources with conflicting priorities. If I pick wrong, a repair line can go down, and downtime costs are 5x higher than normal. I need help to consistently pick the right orders first."

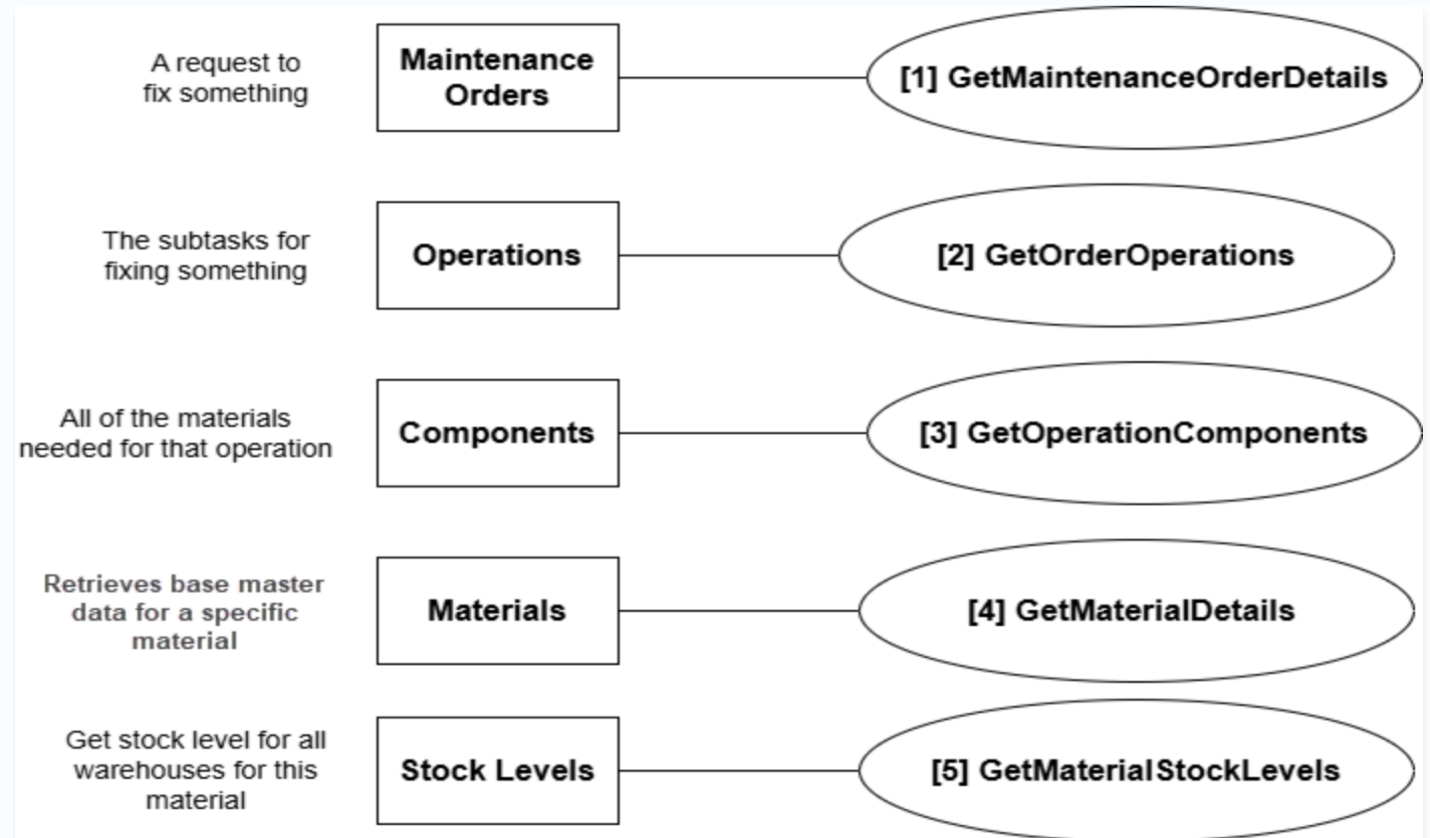
Solution Approach:

- Develop a **CAP-based application** to expose the maintenance order entity and create services for retrieving maintenance order information
- Consume the service as a **destination on SAP BTP**, enabling Joule Agents to securely consume maintenance order data.
- Build a **custom Joule Agent** to retrieve key maintenance order details, including:
 - Order description
 - Status
 - Priority
 - Location
 - Scheduled start and end dates
 - Assigned work center
- Enable a **digital Joule assistant experience** where supervisors can ask questions using natural language and instantly receive relevant maintenance order information.
- Leverage **SAP BDC** to model maintenance order data and build a dashboard that provides maintenance planners with actionable insights.

Business Outcome

As a maintenance supervisor, I want an agent that ranks and re-prioritizes open maintenance orders using risk, cost-of-downtime, and operational context so that I minimize repair-line shutdowns and overall maintenance spend.

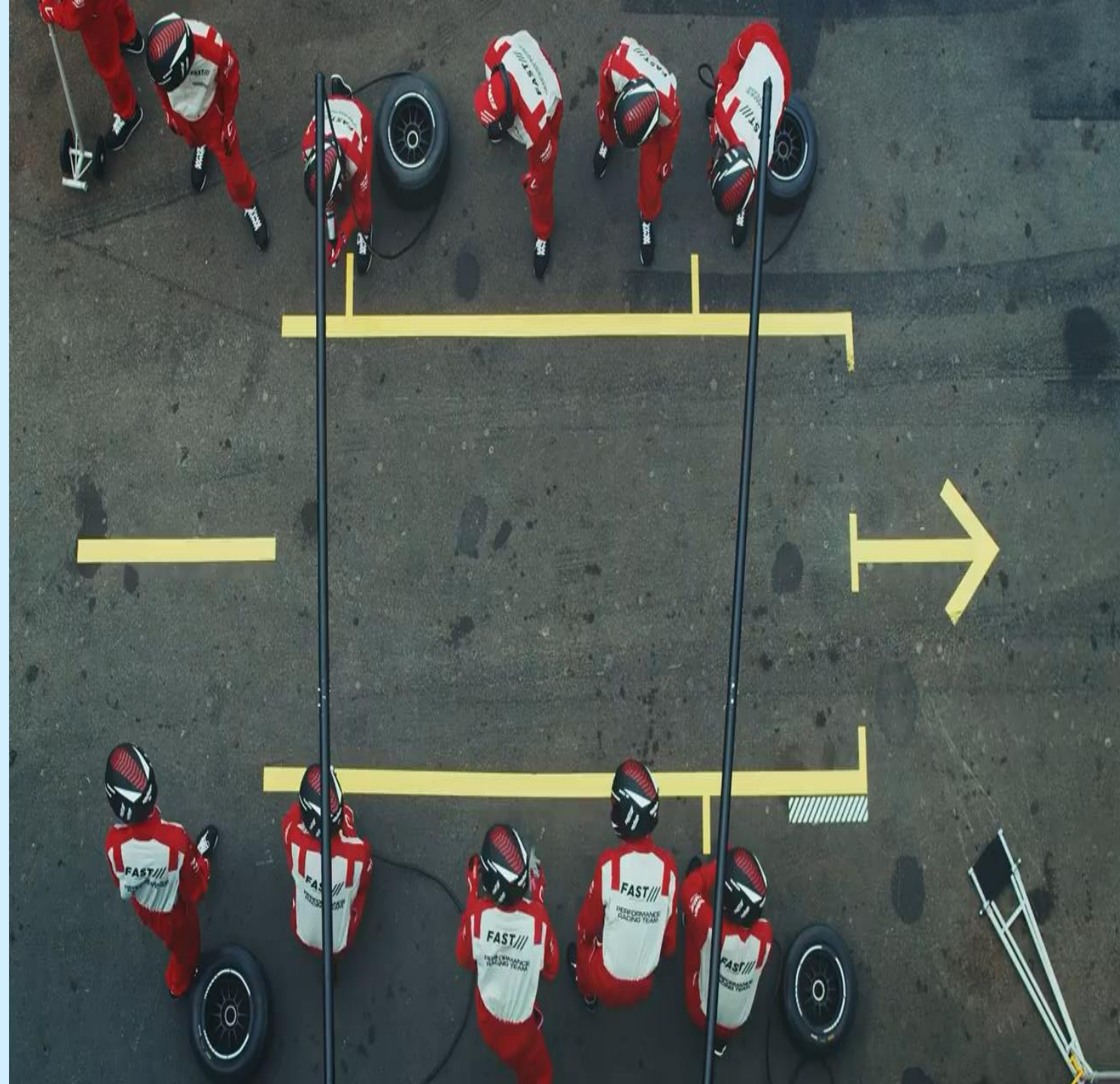
- CAP service on SAP BTP contains the business logic and data access
- Agent Builder in Joule Studio is used to build a custom Joule agent that leverage tools such as Joule skills.
- Joule skills invoke backend actions exposed via destinations
- Conversational interface acts as the entry point and surfaces backend results back to maintenance planners

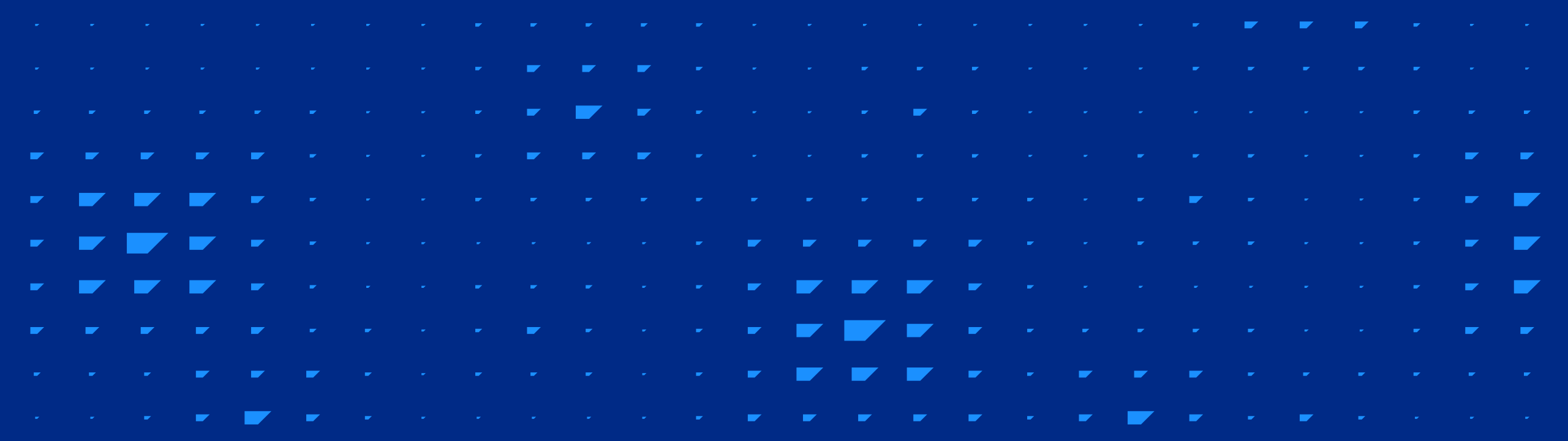


Your perspective matters

Join the conversation

You ask. We answer.





Thank you.

Contact information:

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 **Bring out your best.**